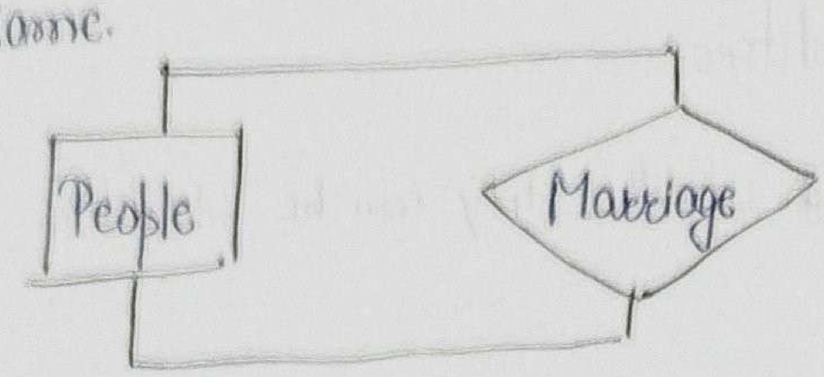
* Degree of Relation: no. of entity sets connected through that relation.

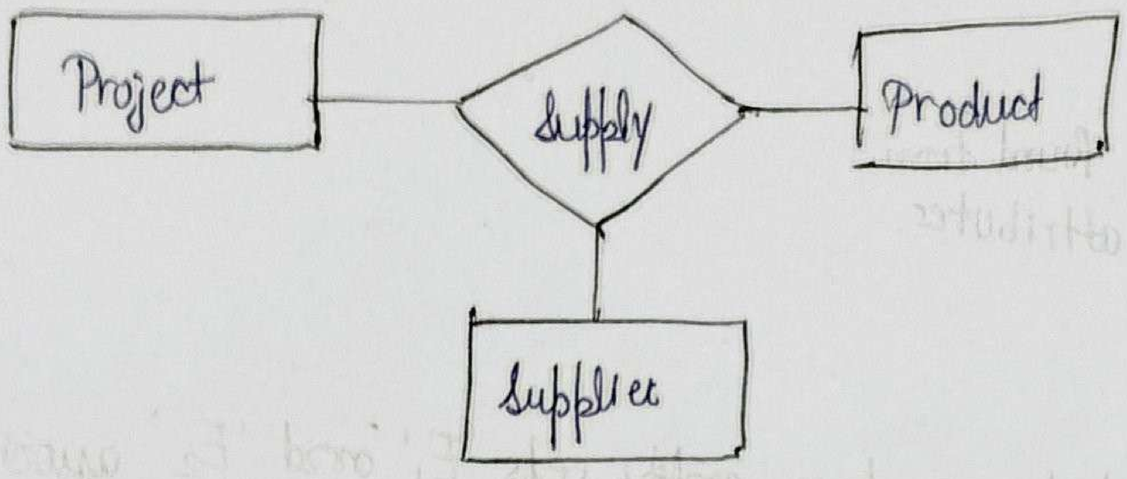
(I) Unary Relation — associates one entity of an entity set to another entity of the same.



(II) Binary Relation — associates one entity of an entity-set 'E_1' to an entity of another entity set 'E_2'



(III) Ternary Relation — associates entities between three entity sets.



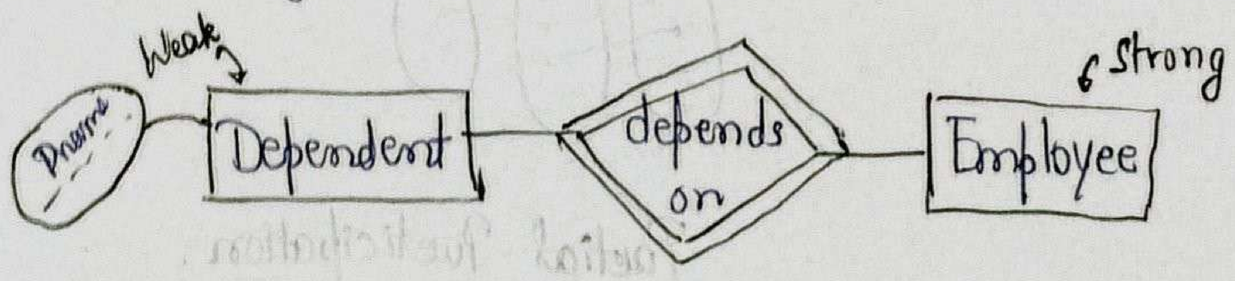
* Domain of an attribute: range of attributes acceptable for that particular attribute

* Role of a relation: name of the relationship (what it is doing)

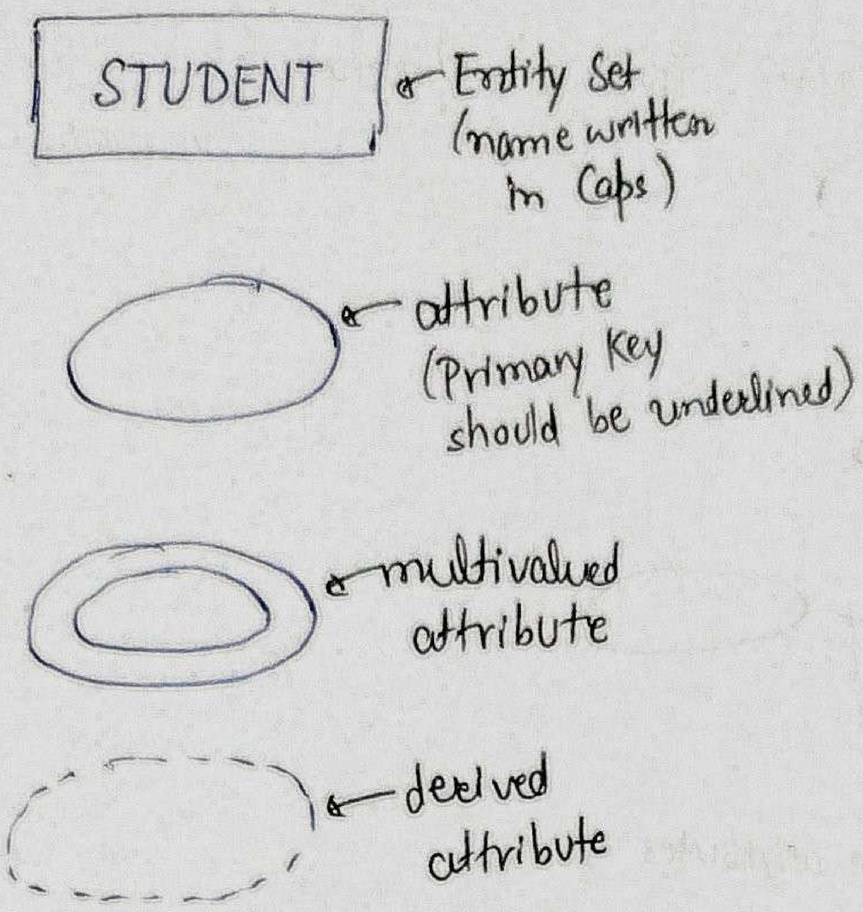
* Cardinality Ratio: no. of entities connected from entity set 'E_1' to no. of entities connected to that of entity set 'E_2' through relation 'R'

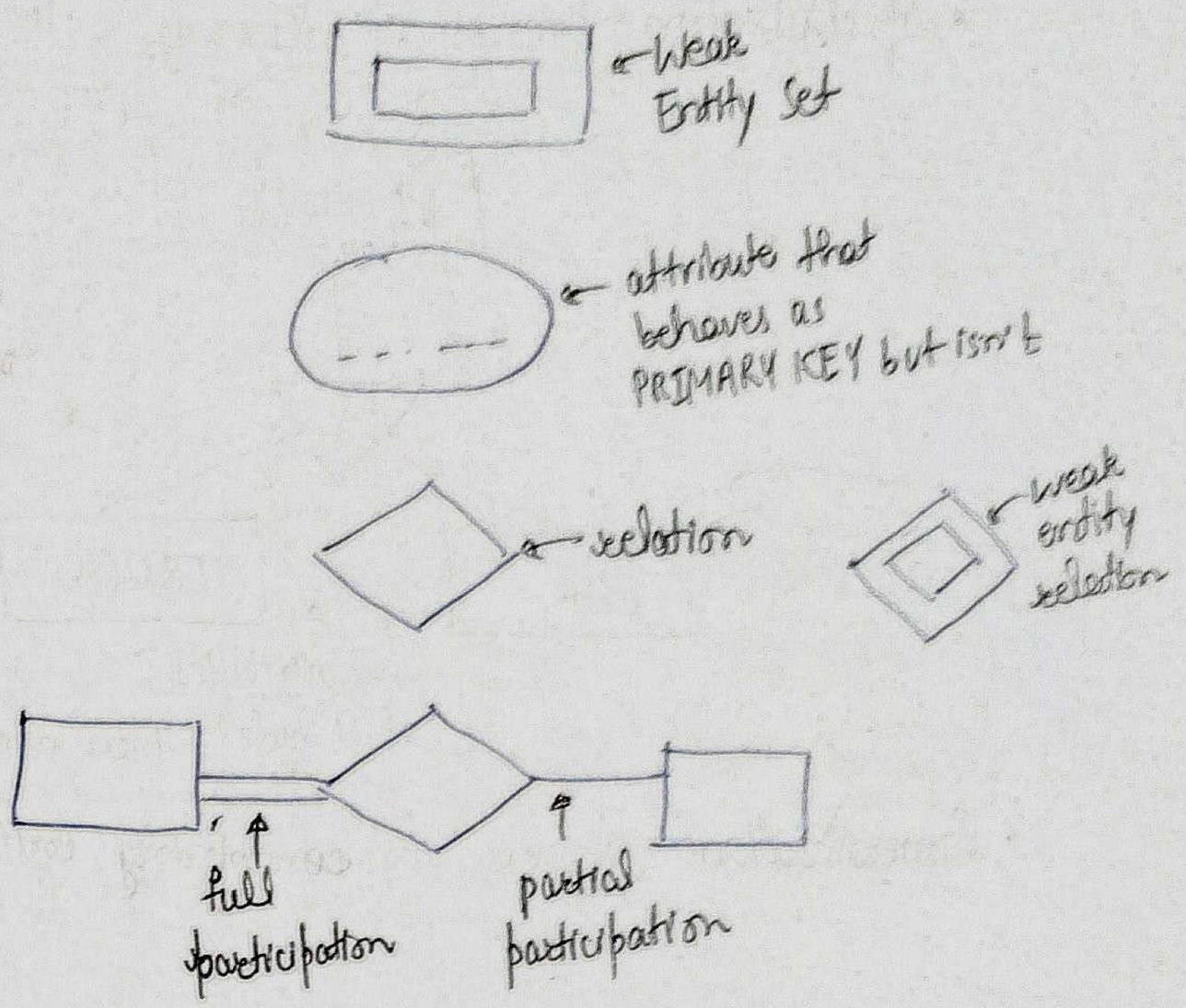
  - (I) 1:1 (one-one)
  - (II) M:N (many-many)
  - (III) M:1 (many-one)
  - (IV) 1:M (one-many)

* Weak Entity Set: can't be uniquely identified by its own attributes alone, and depends on a related strong entity set for identification (no PRIMARY KEY)



Entity Relationship (ER) Diagram





(1) List of entities and attributes

AUTHOR {authorid, name, address}

BOOKS {__}

PUBLISHER {__}

REVIEWER {__}

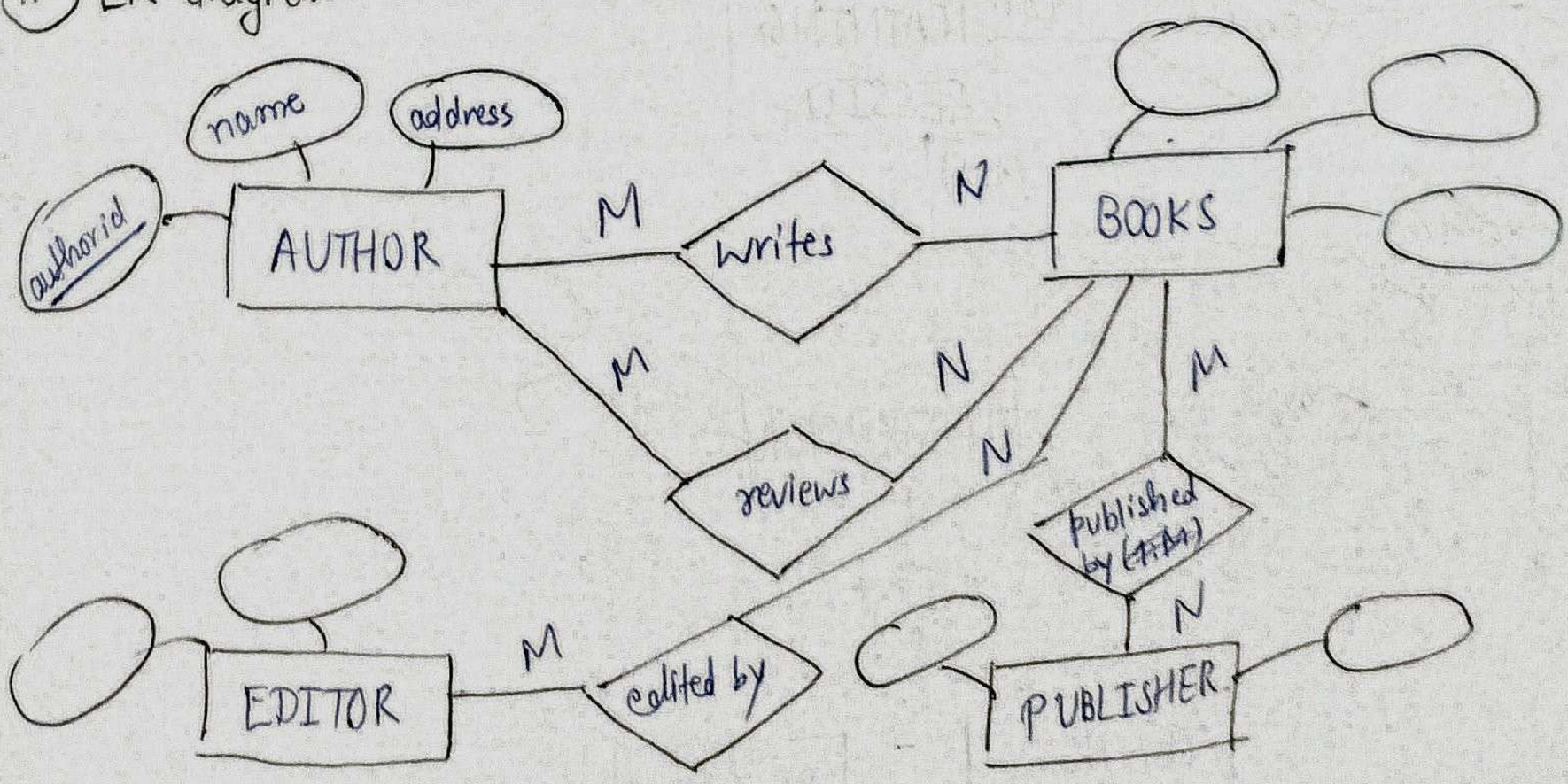
(II) Relationship

AUTHOR → writes → Book (M:N)

AUTHOR → reviews → Book (M:N)

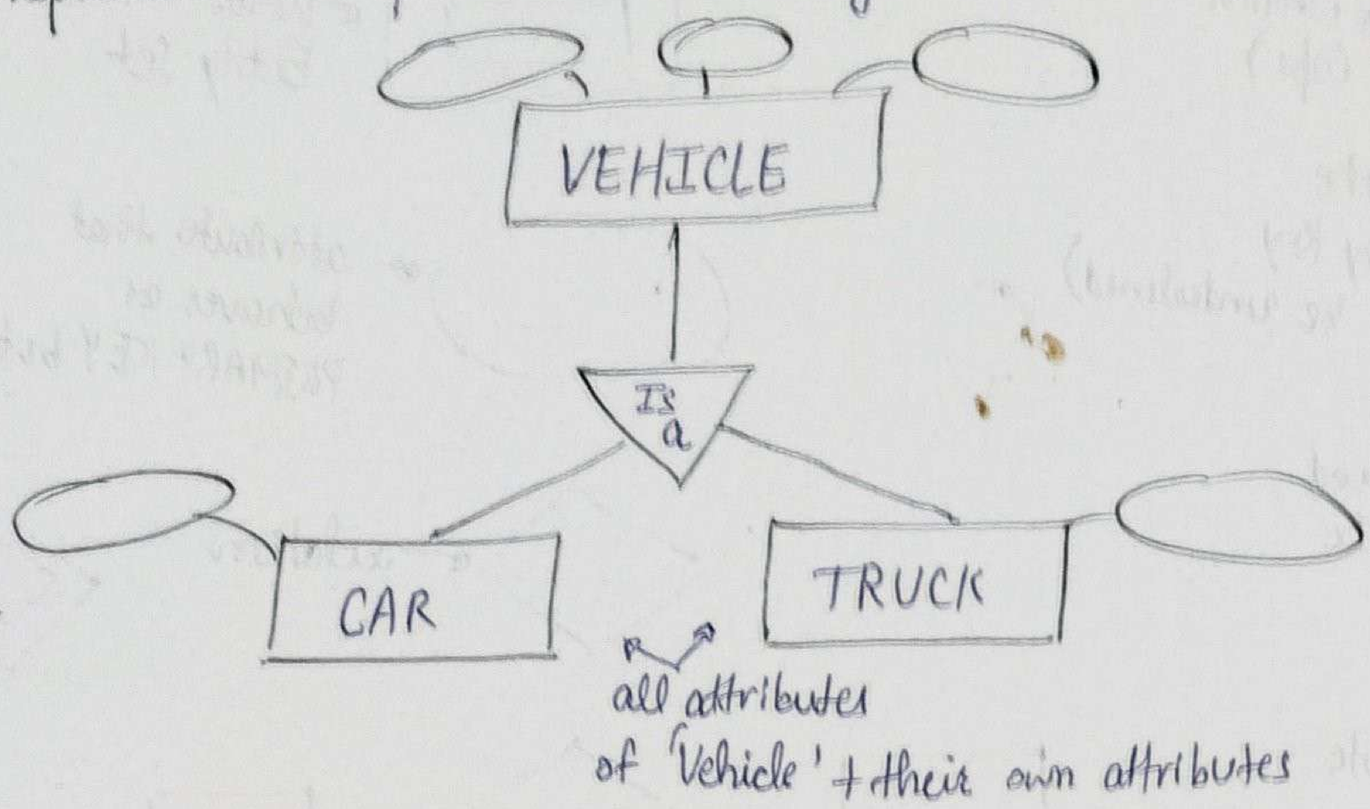
:

(III) ER diagram



* Extended ER diagram (EER): also represents hierarchies

  • Specialization – process of defining subclasses from a superclass (sub-entities from an entity)

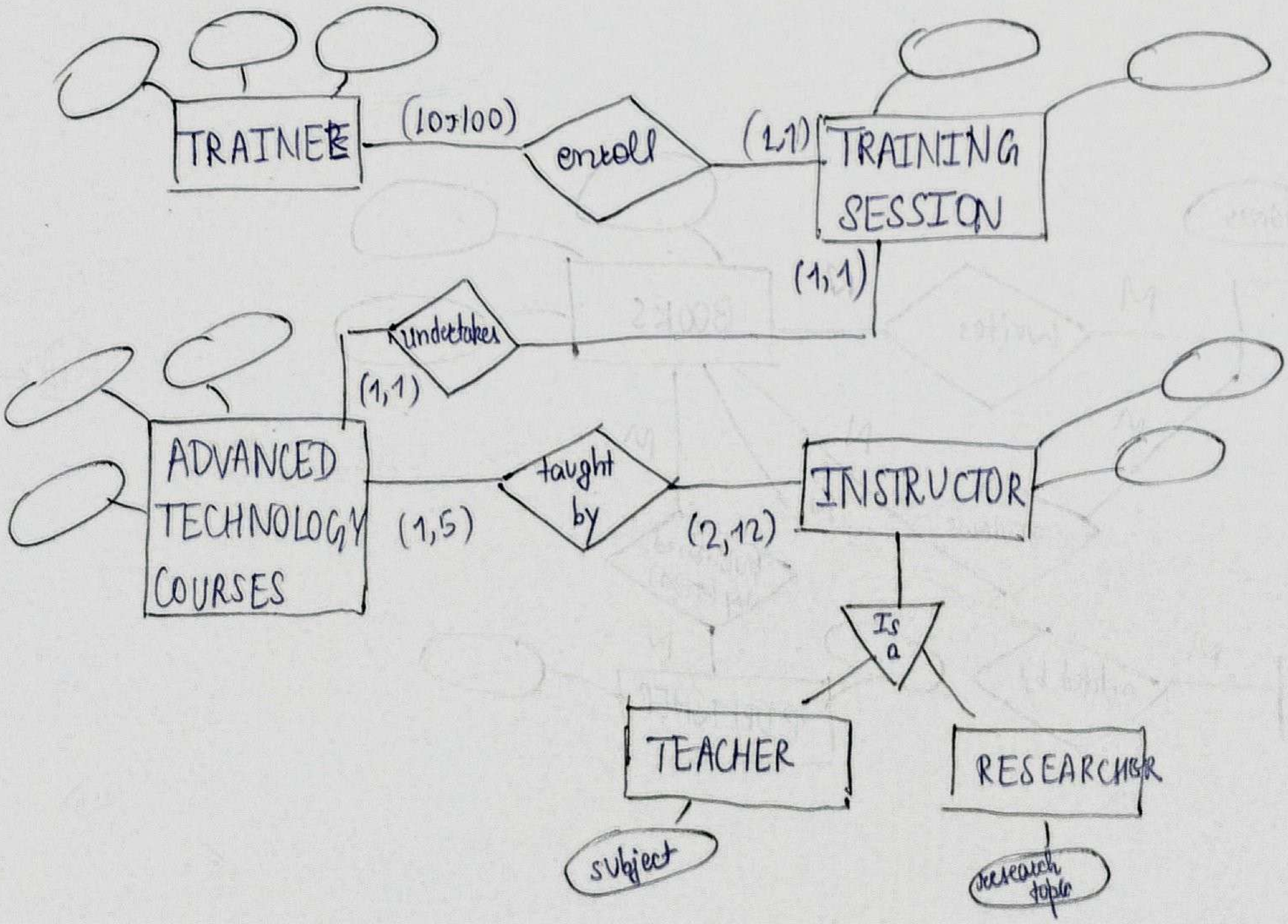


  • Generalization – process of combining entities into a generalized superclass

Case Study:

Design ER diagram for IT training group DB that'll need the i/f for its training program. Clearly indicate the entities, relationships and key constraints. The description of the environment is as follows:

the company has 12 instructors and can handle upto 100 trainees for each training session, the company offers 5 advanced technology courses – each of which is taught by a team of 2 or more instructors, each instructor is assigned to a max^m of 2 teaching teams or maybe assigned to do research, each trainee undertakes 1 advanced tech course per training session



Design ER diagram for a company DB as per the requirements given below. Make appropriate assumptions to complete the specifications.

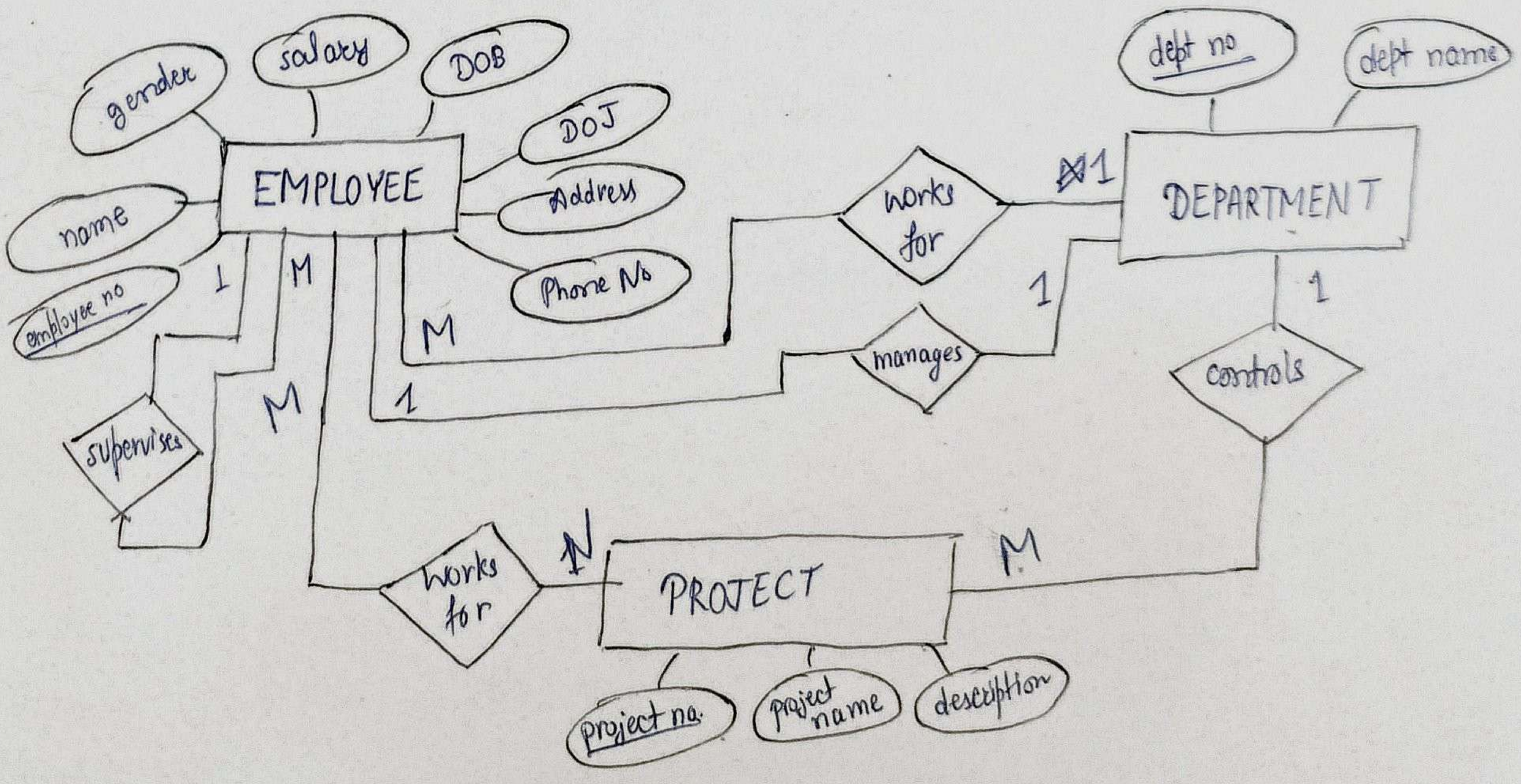
(I) The company stores the i/f about the current working employees. The i/f includes employee no., name, gender, salary and DOB, Date of joining, address and phone no. Each employee works for a dept. for a particular project for a particular no. of hours.

(II) The i/f about dept includes dept. no, dept name. Each dept. controls some project currently running in the company. Also each dept. is managed by a particular employee who becomes the managers for the dept. These employees also supervise all other employees in that dept.

(III) The project i/f includes project no., project name, desceiption

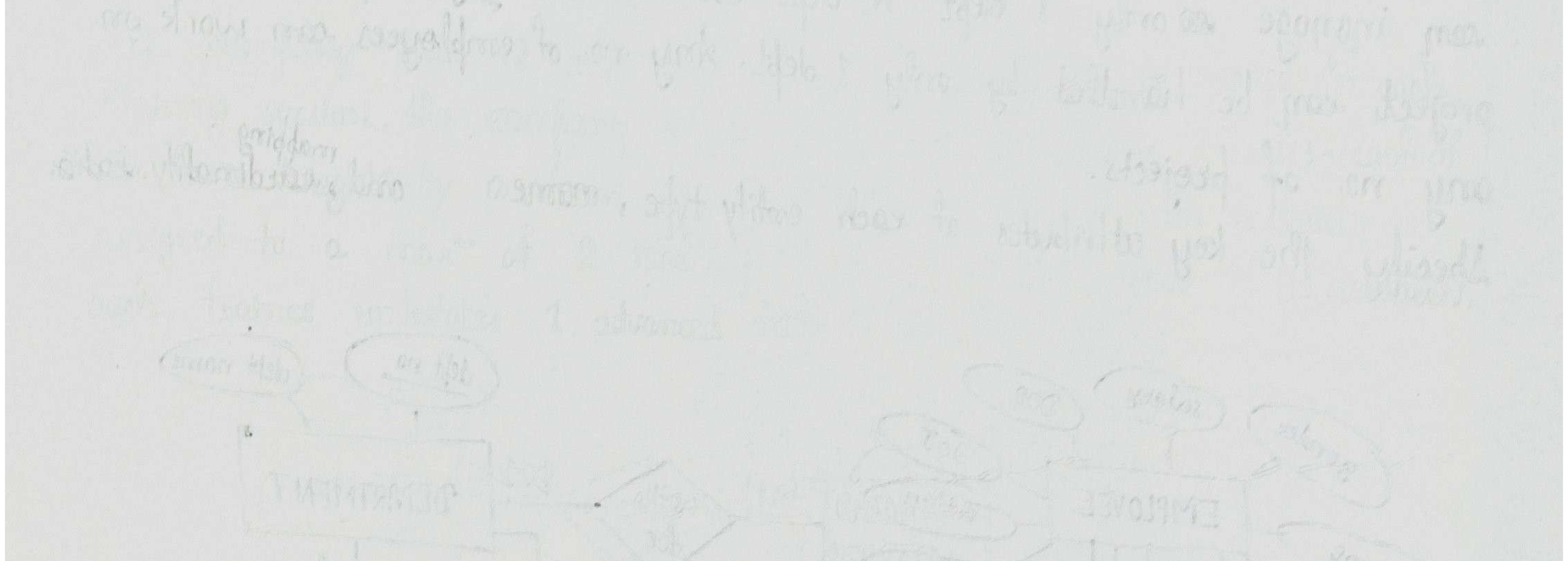
(IV) An employee can only work for only 1 dept, however a dept. can have any no. of employees. A dept. is managed by only 1 manager and a manager can manage only 1 dept. A dept. can control many projects, however 1 project can be handled by only 1 dept. Any no. of employees can work on any no. of projects.

Specify the key attributes of each entity type, name and mapping cardinality.



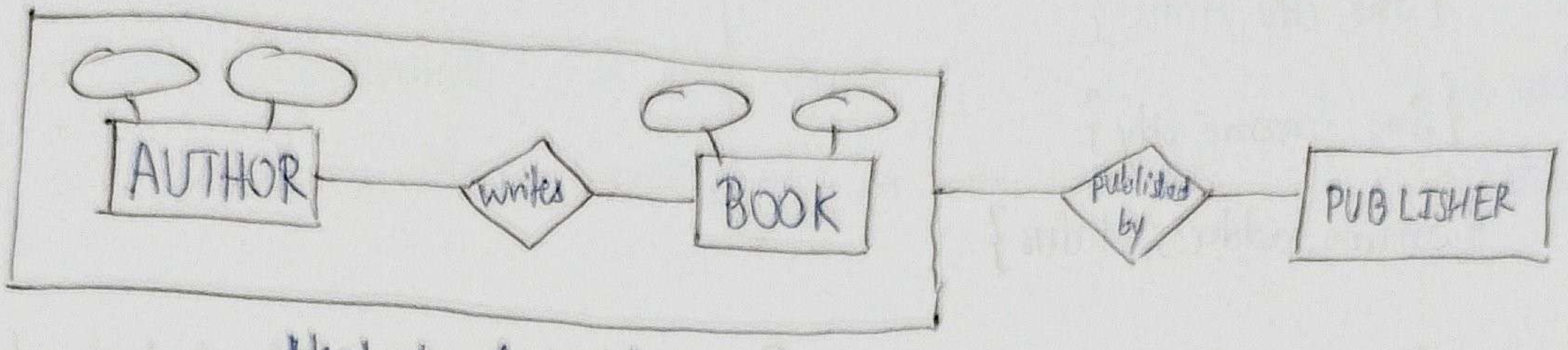
Case Study:

Consider a bank DB gateways having customer, loan, account, employee and branch as entity types. Each branch allows customers to open accounts and borrow loans. A customer can open more than one account and one account may also belong to one or more customers. Similarly, a customer can takeout more than one loan and a loan may be held by more than one customer. The bank has a no. of employees working in different branches of the bank. Add appropriate attributes to each entity type. Represent the key attribute, weak entity type and cardinality ratios. Make appropriate assumptions to complete the specification. Design ER diagram for bank DB.



Aggregation: considering two or more entities and their relationships as a higher-level entity

- the process to ~~represent~~ consider high-level entity constituting of two or more entities and their relationships.



High-level entity

Relational Model

- Relation: representing attributes and values in tabular form
- Relational DB: collection of similar relations.

* Degree of Relation: no. of attributes (columns)
* Cardinality of Relation: no. of tuples (rows)

* Relational Database Instance: at a particular time (instant), the state of the DB.

Keys:

(I) Super Key — set of attributes through which one tuple can be uniquely identified

(II) Candidate Key — minimal subset of Super Key through which a tuple is uniquely identified

(III) Primary Key — the candidate key chosen to uniquely identify each tuple (NOT NULL + UNIQUE)

(IV) Unique Key — takes unique value but can also take NULL values

Characteristics of Relational Model:

- Order of Tuples
- Ordering values of tuples
- Unknown values for any attribute — NULL
- No two tuples are same

(V) Partial Key — in weak entity set, ~~one~~ attribute that behaves like Primary Key

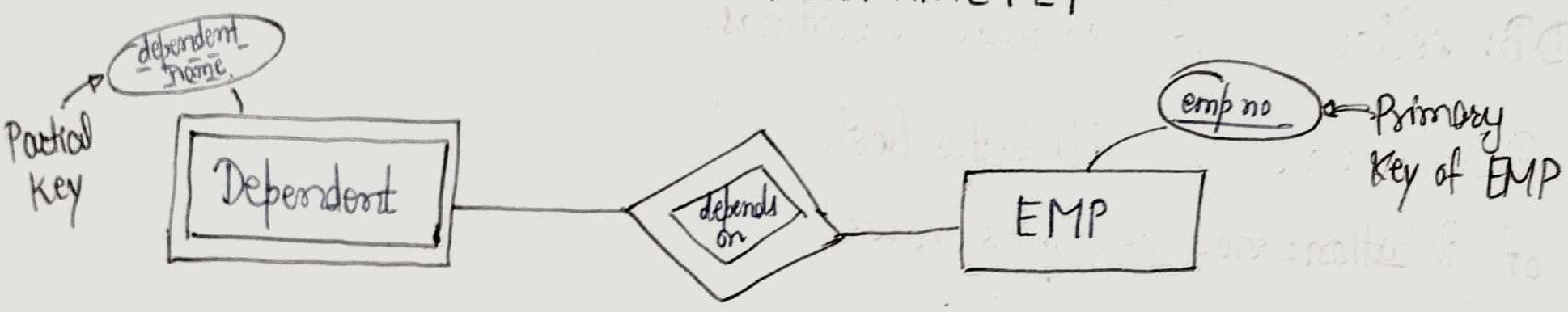
SUP (Sno, Sname, city, status, address, date)

{Sno, city}
{Sno, Sname, address}
{Sno, city, status}
{Sno, Sname, city}
{Sname, address, status}
} Super keys

{Sno}, {Sname, address} } Candidate keys (minimal subset of super keys)

{Sno} ← PRIMARY KEY

{Sname, address} ← ALTERNATE KEY



(VI) FOREIGN KEY — the attribute of one relation can be accessed in another relation by enforcing a link b/w the attributes of two relations.

- A relation that references another relation is known as "Referencing relation"
- A relation that is being referenced is known as "Referenced relation"

EMP

Eno	Ename	Address	Dno
10	aaa	Patna	1A
11	bbb	Delhi	1A
12	ccc	Mumbai	1B
13	ddd	Kolkata	1D

1D → inconsistency

DEPT

Dno	Dname
1A	B.Tech
1B	BCA
1C	MCA

Dno → PRIMARY KEY of DEPT
FOREIGN KEY of EMP

* Self-referencing foreign key:

EMP

Eno	Ename	Address	Mno
10	— —	— —	12
12	— —	— —	NULL
13	— —	— —	14
14	— —	— —	12

Eno ← Primary key
Mno ← self-referencing foreign key

tgY

(i) Entity Integrity Constraints — all Primary Keys are UNIQUE and NOT NULL

(ii) Domain Integrity Constraint — each attribute should take a range of values called the "domain" of that attribute

(iii) Referential Integrity Constraint — if the domain of an attribute 'S_1' of a relation 'R_1' should be same as the domain of an attribute 'S_2' of a relation 'R_2', if:

$$R_1 \rightarrow S_1 \text{ (P.K)}$$
$$R_2 \rightarrow S_2 \text{ (F.K)}$$

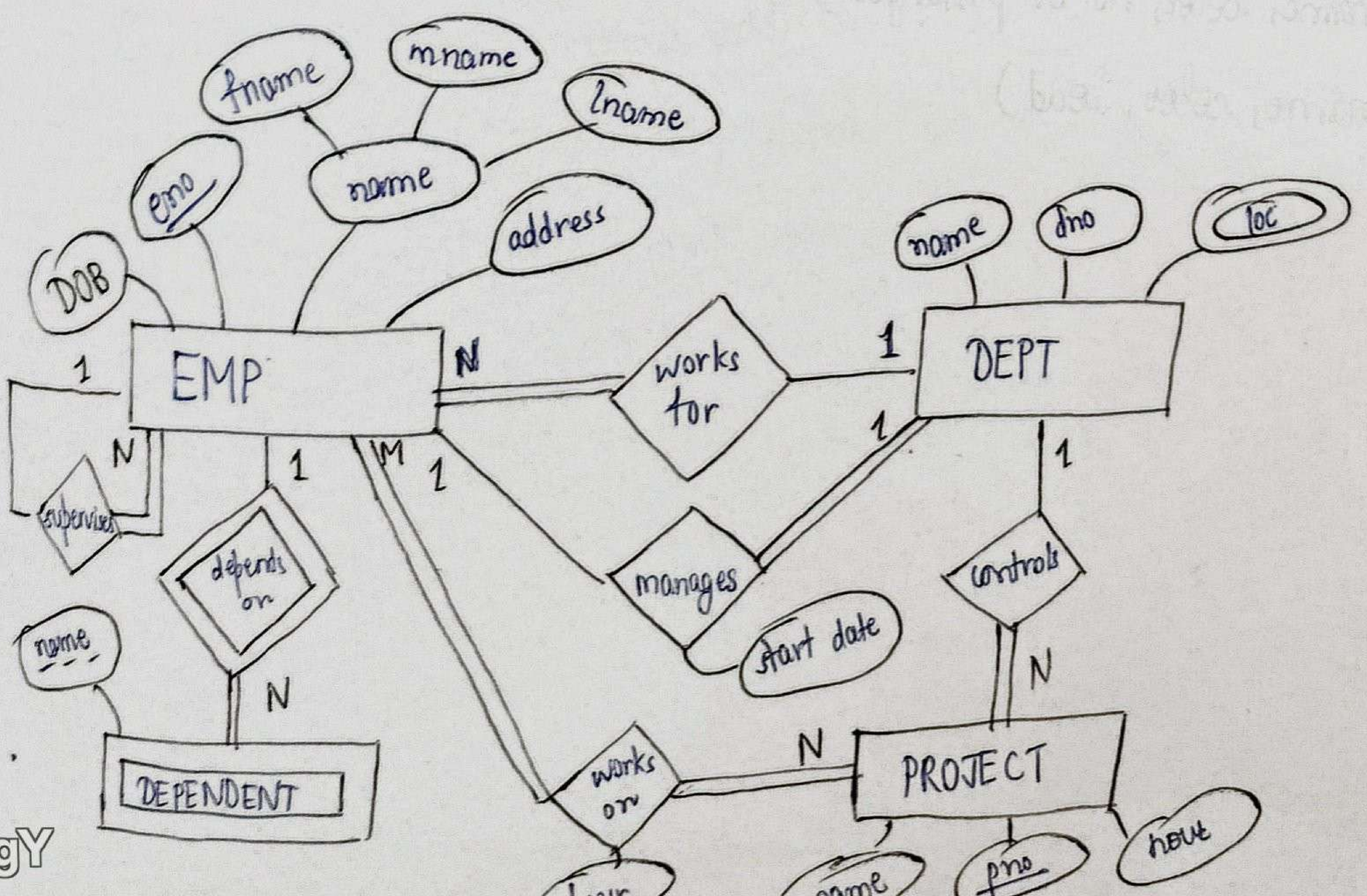
- each tuple value for the F.K should be from the domain of 'S_1', or NULL.

(1) domain for a given set of attributes say 'S_1' in a relation 'R_1' should be the values appear in a certain set of attributes say 'S_2' in the 'R_2' relation the attribute in 'S_1' is same as the attribute in 'S_2' in 'R_2' relation, so we can say it refers to 'R_2'

(ii) A value of foreign key in tuple 'T_1' of the current state, either occurs as a value of primary key for some tuple 'T_2' or it is NULL, so:

$$T_1[\text{F.K.}] = T_2[\text{P.K.}] \;||\; \text{NULL}$$

Conversion of E-R model to Relational model



EMP (DOB, eno, fname, mname, lname, address, dno, sno)
F.K. dno; self F.K. sno

DEPT (name, dno, eno, startdate)

DEPT-location (dno, loc)

DEPENDENT (eno, name)
F.K eno

PROJECT (pno, name, hour, F.K → dno)

WorksOn (eno, pno, hour)

(I) Composite attributes are put as simplest attributes in relational model

(II) For multi-valued attributes, we have a separate table [PK → attribute + PK of original table]

(III) For weak-entity sets, combined attribute of its partial key and primary key of the strong-entity set it is dependent on, acts as PK of that table.

(IV) For 1:1 relations, the part having full participation includes the PK of the other part and any attribute of the relation is also added.

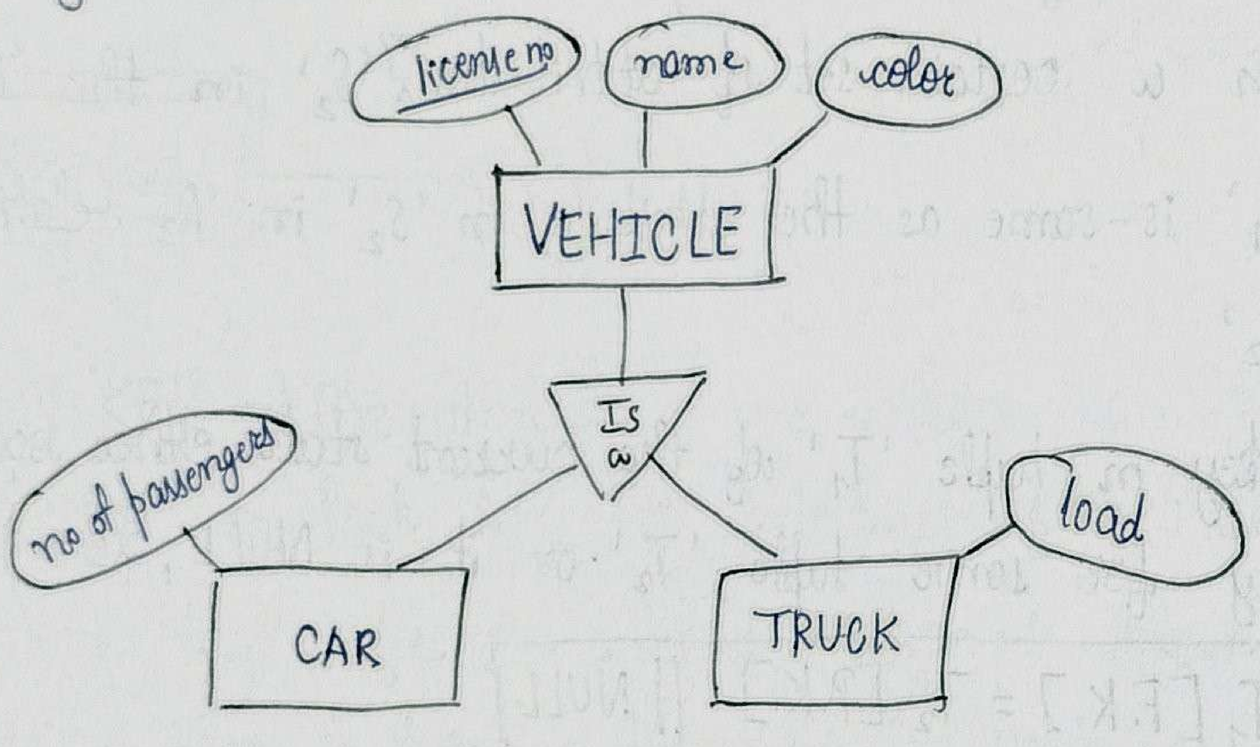
(V) For 1:1 relations, if both sides are full participation, both tables are merged with PK = combined PKs of both tables

(VI) For 1:N relation, the side with 'N' includes the P.K of the side with '1' as F.K.

(VII) For M:N relation, a table is created with P.K = combined P.Ks of both table.

(VIII) For self-referential F.K, a self-referential F.K. is added to that table
(1:N)

(IX) For ternary relation, a table is created with P.K = P.Ks of all three entities' P combi



VEHICLE (license no., name, color)

CAR (license no., name, color, no. of passengers)

TRUCK (license no., name, color, load)